MATHS MANTHAN
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Subject: Mathematics **WEEKLY CHAPTER TEST – 02 (QUADRATIC EQUATIONS)** **Time:** 45 Minutes

Class: Class X (CBSE) **Chapter 4: Quadratic Equations** **MM: 20**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choice is provided in Question 10(iii).
2. **Section A** comprises 4 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 1 VSAQ of 2 marks and 3 SA questions of 3 marks each.
4. **Section C** comprises 1 Applied Case Study with 3 sub-parts of 4 marks total.

SECTION A: Multiple Choice Questions (5 × 1 = 5 Marks)

Q1. Which of the following equations has the sum of its roots equal to 3? [1]
(A) $2x^2 - 3x + 6 = 0$ (B) $-x^2 + 3x - 3 = 0$ (C) $\sqrt{2}x^2 - \frac{3}{\sqrt{2}}x + 1 = 0$ (D) $3x^2 - 3x + 3 = 0$

Q2. Values of k for which the quadratic equation $2x^2 - kx + k = 0$ has equal roots are: [1]
(A) 0 only (B) 4 (C) 8 only (D) $0, 8$

Q3. Which of the following equations has two distinct real roots? [1]
(A) $2x^2 - 3\sqrt{2}x + \frac{9}{4} = 0$ (B) $x^2 + x - 5 = 0$ (C) $x^2 + 3x + 2\sqrt{2} = 0$ (D) $5x^2 - 3x + 1 = 0$

Q4. The quadratic equation $2x^2 - \sqrt{5}x + 1 = 0$ has: [1]
(A) two distinct real roots (B) two equal real roots (C) no real roots (D) more than 2 real roots

Q5. **Assertion (A):** The quadratic equation $x^2 + 2x + 5 = 0$ has no real roots.
Reason (R): The discriminant of $x^2 + 2x + 5 = 0$ is negative ($D = -16 < 0$). [1]
(A) Both A and R are true and R is correct explanation of A. (B) Both are true, but R is not correct explanation.
(C) A is true, R is false. (D) A is false, R is true.

SECTION B: Short Answer Type Questions (Part I)

Q6. Solve for x by factorization: $\sqrt{2}x^2 + 7x + 5\sqrt{2} = 0$. [2]

Q7. The altitude of a right triangle is 7 cm less than its base. If the hypotenuse is 13 cm, find the other two sides. [3]

SECTION B: Short Answer Type Questions (Continued) (Total Sec B = 11 Marks)

Q8. Solve the following quadratic equation for x: [3]

$$\sqrt{3}x^2 + 10x - 8\sqrt{3} = 0$$

Q9. The difference of squares of two numbers is 180. The square of the smaller number is 8 times the larger number. Find the two numbers. [3]

SECTION C: Applied Case Study Based Question (1 × 4 = 4 Marks)

Q10. Case Study: Eco-Tourism River Safari & Upstream-Downstream Dynamics

An eco-tourism resort operates a motorized safari boat on a river. The manufacturer certifies the speed of the motorboat in still water as 20 km/h. The river flows at a uniform speed of x km/h ($x < 20$). The boat takes tourists 48 km upstream to a scenic gorge and returns back downstream to the base dock in a total time of 5 hours.

Based on the above navigation data, answer the following questions:

(i) Write expressions for the downstream speed and upstream speed of the boat in terms of x. [1]

(ii) Formulate the quadratic equation in x in standard form representing the round-trip duration. [1]

(iii) Determine the speed of the river current (x). [2]

OR

Find the time taken for the upstream journey and downstream journey separately. [2]

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